$$1 = \frac{b^2}{r_{min}^2} + z\frac{r_{min}^2 b}{R^3} - 3z\frac{b}{R}, \tag{34}$$

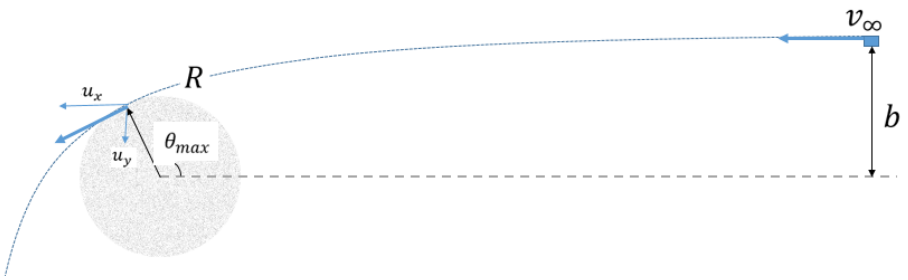
$$r_{min} = \sqrt{\frac{(\frac{3zb}{R}+1) \pm \sqrt{(\frac{3zb}{R}+1)^2 - \frac{4zb^3}{R^3}}}{2\frac{zb}{R^3}}}. \tag{35}$$

$r_{min} = 0$ $b = 0$.

$$r_{min} = R\sqrt{\frac{\left(\frac{3GM}{v_\infty^2 R}+1\right) - \sqrt{(\frac{3GM}{v_\infty^2 R}+1)^2 - \frac{4b^2 GM}{R^3 v_\infty^2}}}{\frac{2GM}{v_\infty^2 R}}} = 4.97 \times 10^9 m. \tag{36}$$

2.3 **[1.0점]** $v_{\infty,min}$.

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$$\frac{mv_{\infty,min}^2}{2} = \frac{m0^2}{2} + \frac{mu_\tau^2}{2} - G\frac{Mm}{R}, \tag{37}$$

$$u_\tau R = v_\infty b, \tag{38}$$

$$v_{\infty,min} = \sqrt{\frac{2GM}{R\left(\frac{b^2}{R^2}-1\right)}} = 252\ km/s. \tag{39}$$

2.4 **[0.6점]** Δr

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r

$$\Delta A = G\frac{(\rho_4\frac{4}{3}\pi r^3)(\rho_4 4\pi r^2 \Delta r)}{r} = \frac{16}{3}\pi^2 G\rho_4^2 r^4 \Delta r, \tag{40}$$

$$A = \frac{16}{3}\pi^2 G\rho_4^2 \int_0^R r^4 \Delta r = \frac{16}{15}\pi^2 G\rho_4^2 R^5 = 1.33 \times 10^{45} J. \tag{41}$$

	평가 항목	배점	
1.1	(1): $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}^2}$ $F = G\frac{m_1 m_2}{r_{12}^2}$.	0.2	**0.7**
	(2) $E = \frac{\sigma}{2\varepsilon_0}$	0.2	
	(4) $h = \frac{g_1}{2\pi G\rho_1}$	0.2	
	$h = 78.0km$	0.1	
1.2	(5) $\Omega_1 = 2\pi$	0.2	**0.5**
	(6) $\alpha = \frac{g_1}{2\pi}$ $\alpha = G\rho_1 h$	0.2	
	$\alpha = 1.56 \times 10^{-2} m/s^2$	0.1	